

Software Implementation and Testing Document

For

Group 27

Version 1.0

Authors:

Carter Mason
Christopher Ko
Makhtar Ndiaye
Nathaniel Arteaga
Talia Lawrence

1. Programming Languages (5 points)

The programming languages utilized in the project right now is simply just JavaScript; utilizing both express js and react js in the environment.

2. Platforms, APIs, Databases, and other technologies used (5 points)

We are planning to use MongoDB as our database to hold persisting information within the project as well as creating the front-end with utilities such as tailwinds CSS and utilizing react and JSX for front-end styling and precise virtual dom rendering control.

3. Execution-based Functional Testing (10 points)

For testing, we will do unit tests and integration testing to make sure the individual components and interactions work as expected. We will use Jest for testing the JavaScript and utilized postman to send faux API requests to our express.js to ensure the proper data is being loaded and displayed properly.

4. Execution-based Non-Functional Testing (10 points)

We will conduct performance testing to test the responsiveness of the application under different workloads. This will include using Lighthouse to analyze page load speeds, accessibility, and overall optimization. Additionally, we will run security tests by validating user authentication, role-based access control, and encryption techniques to ensure data protection. Cross-browser testing will also be performed to confirm that the application functions consistently across different web environments.

5. Non-Execution-based Testing (10 points)

We want to do code reviews and static code analysis to identify and potential inefficiencies or inconsistencies in our implementation. ESLint will be used to enforce coding standards and catch syntax errors early in the development process. Additionally, we will conduct walkthroughs where team members review the logic and structure of the codebase to ensure clarity and maintainability before deployment.